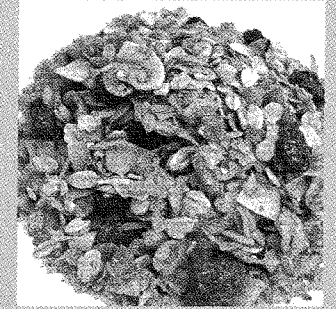
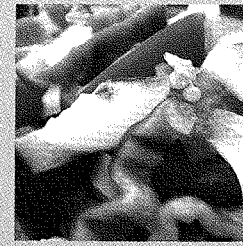




Medical Nutrition Therapy Guidelines

for Hyperlipidemia

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Acknowledgement

We would like to record our full appreciation to the following company and institution for their contributions in kind and facilities.

- Institut Jantung Negara Sdn Bhd - for the use of meeting rooms and refreshments
- The Medical Peer Reviewers
- Abbott Laboratories Malaysia Sdn Bhd - for their services



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Appendix 1: Essential Patient Education Topics for Patients With
Cardiovascular Disease

Appendix 2: Risk Factors for Coronary Heart Disease

Appendix 3: High Saturated Fat & Cholesterol Food List

Appendix 4: Sample 1800 Calories Menu -Low Cholesterol, Low Fat

Appendix 5: Medication

INTRODUCTION

Cardiovascular disease (CVD) is the leading cause of death in Malaysia. Of the CVDs, coronary heart disease (CHD) is the most prevalent cause of death, followed by stroke. Patients with hyperlipidemia are at high risk for a heart attack. Several studies such as the Scandinavian Simvastatin Survival Study (4S), the Cholesterol and Recurrent Events (CARE), Long Term Intervention with Pravastatin in Ischemic Disease with lipid lowering diet (LIPID) showed that lowering the total cholesterol and LDL cholesterol decreased coronary heart disease (CHD) related deaths. Blood cholesterol level is affected not only by what one eats but also by how quickly the body makes LDL cholesterol and disposes of it. These factors determine whether the LDL cholesterol level is high or low. The factors include heredity or family history, what one eats, weight, physical activity and exercise, age and sex, alcohol and stress. Risk factors such as age over 45 for men and over 55 for women, smoking, diabetes, obesity / overweight, high blood pressure and low HDL levels further increase the risk in these patients.

In the USA, the rate of CVD and stroke has declined in men but has increased slightly in women. Computer modeling studies have shown that 25 % of the decline in CHD is attributable to primary prevention and 70% to behavioral changes affecting risk factors or improvements in treatment (Hunink et al 1997). The reasons for the decrease in mortality include better treatment and preventive efforts such as lifestyle changes to modify risk factors. Medical nutrition therapy, which includes physical activity, is the primary intervention for patients with elevated LDL cholesterol. Medical nutrition therapy is critical for secondary prevention because saturated fatty acid levels are related to disease progression in men (Watts et al, 1996).

The objectives of this Medical Nutrition Therapy (MNT) guideline for hyperlipidemia are:

- To standardize dietary management among dietitians.
- To provide guidelines for improving the lipid profile of hyperlipidemic patients.
- To effectively assist clinical dietitians and health care providers to manage hyperlipidemic patients and prevent or delay complications.
- To provide updated knowledge and evidence based Medical Nutrition Therapy.

Treatment strategies have been based on levels of evidence using the following Ministry of Health (MOH) system:

GRADE A	At least one meta-analysis, systemic review, or randomized controlled trial, or evidence rated as good and directly applicable to the target population
GRADE B	Evidence from well-conducted clinical trials, directly applicable to the target population, and demonstrating overall consistency of results; or evidence extrapolated from meta-analysis, systemic review, or randomized controlled trial
GRADE C	Evidence from expert committee reports, or opinions and/or clinical experiences of respected authorities; indicates absence of directly applicable clinical studies of good quality.

OBJECTIVE OF NUTRITION MANAGEMENT

Management of Medical Nutrition Therapy which includes physical activity is the primary intervention for patients with elevated LDL cholesterol. Diet, exercise, and weight reduction is shown to reduce serum lipid levels in patients. Previously, the ATP dietary changes were divided into 2 levels: NCEP Step I and Step II diet. The Step I diet was more liberal in SFA (up to 10 % energy) and dietary cholesterol (up to 300 mg daily) than the Step II diet (less than 8% energy as SFA) and cholesterol (up to 200 mg daily).

Now the ATP III dietary recommendations or known as Therapeutic Lifestyle Change (TLC) diet, are an enhanced Step II diet. SFA recommendations are lowered to 7% of the calories and total fat contents are changed from less than 30 % of energy to a range of 25 % to 35 % of total calories. Consuming 30 to 35 % calories from fat while maintaining a low SFA and trans fatty acid intake is recommended for individuals with metabolic syndrome. This higher fat intake emphasizing PUFA and MUFA, can be beneficial in lowering triglycerides and raising HDL cholesterol with metabolic syndrome.

With a more liberal fat intake, LDL-C can be lowered without exacerbating blood glucose level. The time course for Medical Nutrition Therapy is 3 - 6 months process. Lowering SFA and cholesterol is the first level of behavior change. The TLC diet is followed for 6 weeks. At the second visit, the LDL response is evaluated, therapy is intensified and dietary compliance monitored.

The primary goal of medical nutrition therapy is to provide a meal plan that supports improvement in lipid levels and risk factor management of cardiovascular disease. Current evidence-based guidelines recommend nutrition therapy as the first line therapy for management of dyslipidemia (ATP III). For this purpose, the dietitian plays an active role in assisting the interdisciplinary team in designing the optimal nutrition prescription that coincides with pharmacotherapy when necessary.

Expected outcomes of MNT are:

- Progressive reduction in total fat, saturated fat and cholesterol levels.
- Achievable, desirable body mass index and waist circumference. (ATP III; NIH, 1998).

Major independent risk factors (exclusive of LDL cholesterol) used to predict the 10-year risks for coronary heart disease are based on the Framingham risk evaluation scores. Risk factors include:

- Cigarette smoking
- Hypertension (blood pressure higher than 140/90 mm Hg and / or receiving antihypertensive medications)
- Low level of HDL cholesterol (less than 1.0 mmol/l for males and less than 1.3 mmol/l for females)
- Family history of premature CHD (CHD in male first- degree relative under 55 years of age; CHD in female first-degree relative younger than 65 years)
- Age (men 45 years or older; women 55 years or older)

NUTRITIONAL ASSESSMENT

Initial Assessment

Component	Assessments
Anthropometric Data	- Weight, height and BMI - Weight history - Waist Circumference
Biochemical Data	- Obtain: - Fasting Serum Lipid profile: - Total Cholesterol (TC) - Low Density Lipoprotein Cholesterol (LDL-C) - High Density Lipoprotein Cholesterol (HDL-C) - Triglycerides (TG) - TC/HDL ratio - Fasting glucose level - Renal profile - Liver Function Test
Clinical Data	- Determine types of hyperlipidemia (refer special topics) - Blood pressure - Current medication and drug history - Medical/family history of risk factors - Overall health status and prognosis
Dietary Evaluation	- Diet history: - Previous and current diet practices - Appetite and tolerance of oral intake - Food allergies - Complete nutrition history (using one or a combination of the 24 hour recall / usual food intake / food frequency and pattern of intake according to the disease) - Others: - Types of cooking oil used - Dietary supplements and special food products - Smoking and alcohol intake - Frequency of eating out and food choices
Physical Activity	- Determine: - Level of physical activity and recent changes in exercise habits - Level of cardiovascular activities such as walking, swimming and cycling

NUTRITIONAL ASSESSMENT

Component	Assessments
Psychosocial and Economic Issues	- Assess socio-economic status which includes: - living conditions & family situations - type of employment - financial status - availability of foods - capability of cooking, food preparation and cooking methods - Social, leisure, and travel activities
Knowledge, Attitudes and Motivation	- Assess basic knowledge of cardiovascular disease, types of drugs taken, dietary knowledge of cholesterol and fats - Assess patients' willingness and ability to learn and adapt appropriate changes
Follow-up	- Document agreed short-term and long-term goals in patients' medical records - Follow-up appointment and plans within 4 - 6 weeks (1st follow-up), or 3 - 6 months, whichever is applicable. (Refer to chart on monitoring)

Follow-up Assessment

Component	Assessments
Anthropometric Data	Weight changes
Biochemical Data	Review any new and updated laboratory data
Clinical Data	- Review blood pressure reading - Reported symptoms such as recurrent chest pain, flatulence due to medications, chronic coughing, edema or ascites and others - Drug compliance or changes of medications - Review changes in medical status and recent/planned therapies
Dietary Evaluation	- Assess understanding of initial nutrition information and food/meal plan - Assess changes in patient's food intake (total fats, saturated fat and fiber intake) and appetite - Assess compliance rate
Exercise	Review and reinforce activity level or exercise habits
Follow-up	- Recommended follow-up visits within 3 months if goals not achieved - If goals have been met, recommend ongoing nutrition care within 3 - 6 months

NUTRITION PRESCRIPTION

Nutrition prescription is determined based on nutrition history, treatment modality, treatment goals and concurrent medical conditions.

Nutrient	Recommendation / Rationale	Practical approaches and expected outcomes
Energy/ Calories	Energy recommendation needs to take into consideration weight status, activity level and diet history. Plan total calorie intake to achieve recommended body weight. 30 - 35 kcal/kg/day for normal body weight.	If BMI > 25 kgm ⁻² , patients are strongly recommended to reduce weight by 5 - 10% of the current body weight (ATP III; NIH, 1998; MDA, 1999). Goal: Lose 0.5 to 1 kg per week (ATP III; NIH, 1998; MDA, 1999; USDA 2000). Long-term goal: to achieve and maintain a BMI of < 25 kgm ⁻² .
Carbohydrate	50% - 60% of total energy intake (American Diabetes Assoc, 2003). Lower intake of 50% should be considered for persons with the metabolic syndrome who have elevated TG or low HDL cholesterol (ATP III).	Emphasize on whole grain or complex carbohydrates (at least 3 servings) such as brown rice, wholemeal bread, chapatti, oats, bran, corn, and whole grain breakfast cereals.
Sugar (as part of CHO calories)	High intake of sugar in place of fat is accompanied by a reduction in HDL cholesterol and a rise in triglycerides (Albert et al, 2002; Garg, 1998; Turly et al, 1998). Limit sugar intake to not more than 15% energy.	To identify excessive sugars such as sweets, jam, sweetened drinks, sweetened kulh, condensed milk, cordials, cookies and carbonated drinks.
Starch (as part of CHO calories)	Most of the carbohydrate intake should come from grain products, especially whole grains, vegetables, fruits, as they are high in viscous fiber that helps to lower LDL cholesterol.	Other starches for example white breads, white rice, noodles, roti canai, roti jala, putu mayam, cream crackers and naan should be consumed within the recommended servings.

Nutrient	Recommendation / Rationale	Practical approaches and expected outcomes
Fiber	20 - 30 g fiber per day (of which at least 6 grams should be soluble fiber) (NIH, 1998; MDA, 1999; AHA, 1997; American Dietetic Assoc, 2001; Brown et al, 1999). An increase in soluble or viscous fiber of 5-10g per day is accompanied by an approximately 5% reduction in LDL cholesterol (US Dept of Health & Human Services, 1998).	Fiber should be increased gradually to minimize gastrointestinal discomfort. The recommended minimum amounts of servings are 3 servings of vegetables and 2 servings of fruits plus one serving of wholegrain products. Emphasize on pulses and other whole grain foods to be taken regularly. Soluble fiber is found in oats, oatmeal, oat bran, beans, legumes, barley and citrus fruits. Oats have the highest proportion of soluble fiber among cereals. To get 6 grams of soluble fiber a day you could eat 1/2 cup of oat bran.
Protein	15 % or 0.8 - 1.0 g/kg per day 2 - 3 servings per day.	Fish such as kembong, tenggiri, kurau, siakap, tilapia and keli can be eaten within the recommended servings. Vegetable sources of omega-3 are flaxseed, soya, and canola oil. Choose only lean cuts of meat. Beef/buffalo: sirloin tip, round steak, rump roast, arm roast. Mutton/lamb: shank roast, sirloin roast, loin chop, sirloin chop, and center leg chop. Pork: center-cut ham, loin chops, tenderloin. Other meats: ostrich, deer, frog, rabbit, pigeons, quail (puyuh) can be eaten within the recommended servings. Remove skin and fat from poultry. Avoid wings, feet and other fatty parts.

Nutrient	Recommendation / Rationale	Practical approaches and expected outcomes
Protein	2 eggs per week (MDA, 1999)	Egg yolks should be limited to no more than 2 per week as they contain approximately 215 mg of cholesterol per egg. (Grade C) Omega 3 enriched eggs contains the same amount of fats and cholesterol. As for lower cholesterol eggs, their cholesterol levels have been reduced by 25%. The recommendation for eggs should be based on total intake 200 mg cholesterol per day.
	Replace animal proteins with soy protein whenever possible. Soy protein has been reported to lower LDL cholesterol and total cholesterol (US Dept of Health & Human Services, 1998).	Choose more legumes, pulses, soy and soy products more often to replace animal proteins. Nuts that are high in unsaturated fat can be eaten; 1 oz (30 g) ~ 5 times per week but the intake should fit within the calorie and fat goals. (Grade A)
	Soy protein at least 25g/day lowers LDL levels by 5% (Jenkin et al, 2000) (Grade A) It is recommended to include a variety of protein from both animal and plants sources in the daily diet.	Soy products 25g/day in weekly meal plan. (Grade B)
Dairy products	1 - 2 servings per day	Choose low-fat or non-fat (skim) milk and milk products. This does not include condensed milk, evaporated creamer and non-dairy creamer. Read labels to determine the fat and cholesterol level Reduce high-fat cheese such as Cream cheese, Cheddar, Monterey Jack, Colby, Swiss, American processed, Blue cheese. Look for cheese labeled "low-fat", "light" or "lite".

Nutrient	Recommendation / Rationale	Practical approaches and expected outcomes
Total Fat	25 - 30% of total energy intake (ATP III; MDA, 1999; AHA 2000).	If initial dietary assessment identifies a patient is consuming total fat within 30% of total calories, it is advisable to recommend a lower percentage of total fat. However fat intake should not be lower than 25% of total energy intake since this could involve replacement with excess carbohydrate possibly resulting in higher triglycerides and lower HDL-C. Educate patients on nutrition facts on food labels to compare fat content. For low fat, aim for foods that are less than 3 g of fat per 100 calories.
Saturated fatty acids (SFA)	7 - 10% of energy (MDA, 1999; Ginsberg et al. 1998; Gordon, 1995). Diet high in saturated fats raises serum LDL Cholesterol levels.	SFA raise blood cholesterol levels (ATP III). The main objective is to identify sources of SFA and reduce its intake. Initial diet therapy would be to reduce SFA intake to 10% total energy. Further reduction to 7% of energy may be required if total cholesterol (TC) levels do not improve despite dietary compliance. For a <10% energy SFA, a palm oil with PUFA-rich vegetable oil blend in the ratio of (1:2) may be used as cooking oil. Coconut milk is one of the major contributors to saturated fat intake. Avoid coconut milk. If taken, for home cooking: coconut milk should be limited to 2 tbsp twice a week in any dish. When eating out, coconut milk based gravy is discouraged. Coconut milk, oil and matured coconut flesh are not recommended on a 7% energy SFA-diet. This applies to patients with high LDL cholesterol and triglyceride levels. Low fat santan contains 14 calories, 1.5g of total fat for every one tablespoon compared to 37 calories, 3.8 g of total fat of regular coconut milk Saturated fats such as ghee, butter, shortening, tallow (beef fat), lard (pork fat), palm kernel oil, cocoa butter and full cream dairy products should be minimized in consumption.
Saturated fatty acid (SFA)		

Nutrient	Recommendation / Rationale	Practical approaches and expected outcomes
Polysaturated fatty acids (PUFA)	5 - 7 % of energy intake Fish consumption reduces sudden cardiac death and CHD mortality (Albert et al, 1998; Dolecek et al, 1991; Singh et al, 1997; Kris-Etherton et al, 2002). 2 fatty fish meals per week reduced 50% primary cardiac arrest (Siscovick et al, 1995; Albert et al, 2002).	Most PUFA should be derived from liquid vegetable oils (corn, safflower, sunflower, soy bean, rice bran, sesame) and deep-sea cold-water fish. These oils are not recommended for deep fat frying due to their chemical composition and low smoking point.
Omega-3 fatty acids (alpha-linolenic acid, EPA, DHA)	Omega-3 fatty acids reduces LDL cholesterol levels when substituted for SFA in the diet (Singh et al, 1997; de Lorgeril et al, 1999). Omega-3 fatty acids also lower plasma triglycerides and prevent aggregation of blood platelets (anti-thrombotic effect) - effects which are good for heart health.	Patient with CHD, recommend consumption of more than 2 servings of fish per week considering the benefits observed with intake of omega 3 fatty acids. (Grade B) Sources of omega-3 (EPA + DHA) include deep-sea cold-water fish (herring, salmon, mackerel, sea bass, cod, fresh sardine, and tuna). Alpha-linolenic acid are found in tofu, soybean oil, canola oil, flaxseed and walnuts; these should be taken within the recommended servings
Omega-6 fatty acid (alpha-linoleic acid)		Sources of omega-6 fatty acids Include corn oil, sunflower oil, safflower, soya bean, nuts and seeds
Monounsaturated fatty acids (MUFA)	12-15% energy. When substituted for dietary SFA and trans fats, MUFA have a hypocholesterolemic effect (Mensink & Katan, 1992; Valsta & Jauhiainen, 1992).	Most MUFA should be derived from vegetables sources, including plant oils (canola, olive, peanut) and nuts. These oils are not recommended for deep fat frying.

Nutrient	Recommendation / Rationale	Practical approaches and expected outcomes
Trans fatty acids (TFA)	< 1% energy of total diet (WHO/FAO, 2003). TFA are 2 to 10 times worse than the SF as risk factors of CVD. A	Trans fatty acid are produced mainly by the commercial hydrogenation of PUFA-rich edible oils, a process which "hardens" these oils at a high temperature in the presence of metal catalyst for the purpose of solid-fat food applications (shortening, margarine, vegetable ghee, bakery fat, etc.) The use of liquid vegetable oil, soft margarine made from blended palm oil and trans fatty acid-free margarine are encouraged instead of butter, hard margarine, ghee, hydrogenated vegetable oil and shortening. Reduce intake of products made from partially hydrogenated oils such as baked products include crackers, cookies, doughnuts, instant noodle, snack food, fries, frozen processed foods and pastry.
Dietary Cholesterol	< 300 mg/day (ATP III) High intakes of dietary cholesterol raise LDL cholesterol levels (Schaefer et al, 1997; Schaefer et al, 1995).	Dietary cholesterol is only found in animal products. Cholesterol rich foods that are low in saturated fatty acids namely eggs and to a lesser extent shellfish have smaller effects on LDL-C. The use of no more than 1 - 2 egg yolks per week is allowed inclusive of hidden forms such as in pastries, cakes. If low cholesterol eggs are used, not more than 3 egg yolks per week. However, foods such as internal organs (kidneys, liver, sweetbread, brain, roe), full cream milk, chocolates and ice cream should be minimized. Prawns, squids, crabs and shellfish can be included in the diet plan based on the frequency and amount of intake. Educate patients on nutrition facts on food labels to compare cholesterol content. For low cholesterol, aim for foods that are less than 5% of the Recommended Daily Value.

Nutrient	Recommendation / Rationale	Practical approaches and expected outcomes
Alcohol	< 2 units/day 2 unit (standard serving) alcohol for men and 1 unit for women	Patients with hypertriglyceridemia should avoid alcohol consumption. A unit is defined as 5 ounces of wine, 12 ounces of beer or 1½ ounces of 80-proof whisky. Persons who do not drink should not be encouraged to initiate regular alcohol consumption.
Dietary Sodium	< 2400 mg sodium or 6 g sodium chloride (ATP III; MDA, 1999; JNC VII; AHA 2000).	2400 mg of sodium is equivalent to 1 level teaspoon of salt. Consuming less salt can reduce the chances of developing high blood pressure. High salt intake is associated with higher blood pressure (JNC VII; Chobanian & Hill, 2000).

NUTRITION IMPLEMENTATION

Component	Assessments
Goal setting	- Discuss nutritional / health behavior issues - Individualized short and long term goals - Negotiating priorities
Education	- Discuss basic knowledge on Hyperlipidemia and dietary management of lipid, blood glucose, blood pressure. - Recommendations for life-style changes (e.g. exercise, smoking cessation, weight reduction) - Select appropriate meal planning approach (e.g. Malaysian healthy dietary guidelines, hyperlipidemia guidelines, fat and cholesterol intake) and educational materials. - Educate patient to take responsibility of their disease conditions. - Include caregivers and family members in teaching to provide supportive role.
Meal Planning	- Prioritize the following with the patient: - Regular healthier meals throughout the day, including home cooking and eating out. - Types and quantity of fat taken, - Portions/servings size emphasized. - If alcohol is taken, limit to 2 unit (standard serving) alcohol for men and 1 unit for women per day.
Follow-up	- Review patients for continuing care and monitoring. - Emphasize short and long term nutritional management goals. - Provide new updated knowledge on nutrition in CVD.

Refer to Appendix 1 for detailed patient education topics.

MONITORING

Encounter	Length of Contact	Time Between Encounters
1 2 3 and 4	At least 45 minutes Within 30 minutes Within 30 minutes	- 3 - 6 months As prescribed by physician; reassess dietary compliances and laboratory results in 3 - 6 months

Outcome Assessment Factors	Base-Line Evaluation and/or Intervention at each visit				Expected Outcomes	Ideal/ Goal Value
	1	2	3	4		
Anthropometric Measures*						
Height and Weight, BMI Waist Circumference*	✓	✓	✓	✓	Maintain healthy weight and appropriate waist circumference *	Within reasonable body weight BMI = 18.5 - 24.9 kgm ⁻² Waist Circumference: • Males < 35 inch (90 cm) • Females < 31inch (80 cm) (Grade B)
Laboratory parameters*						
Lipid Profile Total Cholesterol, TC Triglycerides, TG*	✓		✓	✓	↓ TC ↓ TG or no change ↓ Non-HDL (if TG > 2.3mmol/L)* ↓ LDL-C	< 5.2 mmol/L, < 1.7 mmol/L
LDL-Cholesterol						< 4.1 mmol/L (0 - 1 RF) † < 3.4 mmol/L (Multiple 2+ RF) < 2.6 mmol/L (CHD & CHD Risk Equivalent), > 1.1 mmol/L < 6.1 mmol/L
HDL-Cholesterol*					↑ or no change	
Fasting Glucose*					Within normal limit or	
BP*					Within normal limit or ↓	< 130/ <80mmHg, († RF - risk factors. Refer appendix 4 for details).
Therapeutic Lifestyle Changes**						MNT Goal
Nutrient Choice						
Food variety and caloric intake	✓	✓	✓	✓	Selects from beneficial nutrient-rich food sources & plans calorie intake to maintain/reduce weight	Calories to maintain/reduce weight

Outcome Assessment Factors	Base-Line Evaluation and/or Intervention at each visit				Expected Outcomes	Ideal/ Goal Value
	1	2	3	4		
Therapeutic Lifestyle Changes**						
Nutrient Choice						
Fat and cholesterol intake	✓	✓	✓	✓	Limits foods ↑ in cholesterol and saturated fat. Uses monounsaturated fat food sources as the preferred fat	Percentage of total energy intake from fat: • Total Fat = 25-30% - SFA = 7 - 10 % - PUFA up to 4 - 7 % - MUFA = balance • Dietary cholesterol intake of < 200 mg/day (Grade A)
Trans fatty acids intake	✓	✓	✓	✓	Limits foods ↑ in trans fatty acids	Minimal intake of trans fatty acids. (Grade B)
Nuts, fish, soy products intake		✓	✓	✓	Incorporate fatty fish (high in n-3 fatty acids), nuts and soy products in weekly meal plan	Fatty fish (average of 7 oz/ wk) (Grade A), nuts (1 ounce ~5 times/week (Grade A) and soy (25 g/day) products (Grade A) are consistent components of meal plan
Soluble fiber intake		✓	✓	✓	Increases intake of foods ↑ in soluble fiber	Dietary fiber of 20-30 g, 10-25 g of which should be viscous (soluble) fiber (Grade B)
Vitamins/ antioxidants intake	✓	✓	✓	✓	Maintains intake of foods ↑ folate, B-6	Increased intake of: o Folate (RDA 400 mcg/day), B-6 (RDA 1.3mg/day) in foods (Grade B); o Vitamin E (RDA 15 mg/day) in foods (Grade B)
Meal Planning						
Food preparation	✓	✓	✓	✓	Uses cooking techniques to modify fat intake	Consistently uses cooking techniques to modify fat intake
Recipe modification		✓	✓	✓	Modifies recipes to ↓ saturated fat, and trans fatty acids	Recipes consistently modified to ↓ saturated fat, and trans fatty acids
Dining out			✓	✓	Selects foods appropriately from restaurant's menu or at hawker center.	Consistently selects appropriately from restaurant's menu

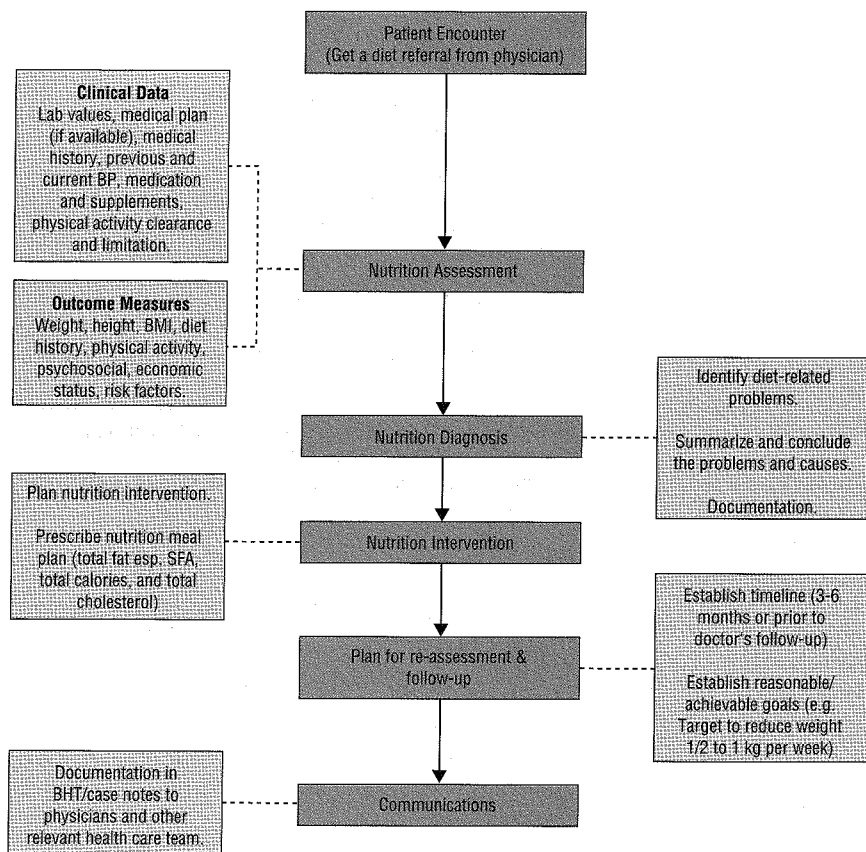
Outcome Assessment Factors	Base-Line Evaluation and/or Intervention at each visit				Expected Outcomes	Ideal/ Goal Value
	1	2	3	4		
Healthful Habits						
Nutrient Choice						
Risks/benefits of alcohol beverages	√	√	√	√	States the risk/ benefits of alcohol beverages	May have up to 1-2 drinks per day of an alcohol containing beverage (Grade C)
Exercise	√	√	√	√	Participates in aerobic activity	Exercise intensity at 40 %-50 % of exercise capacity for ~30 min/day most days of the week (Grade B)
Smoking cessation	√	√	√	√	Verbalizes the importance of smoking cessation	Smoking cessation successful
Medication						
Potential food/drug interaction	√	√	√	√	Verbalizes the potential of food/ drug interaction	No food/ drug interaction

DOCUMENTATION

Documentation may be in SOAP (Subjective, Objective, Assessment, Plan) format or in other style but essential components of documentation should include the following:

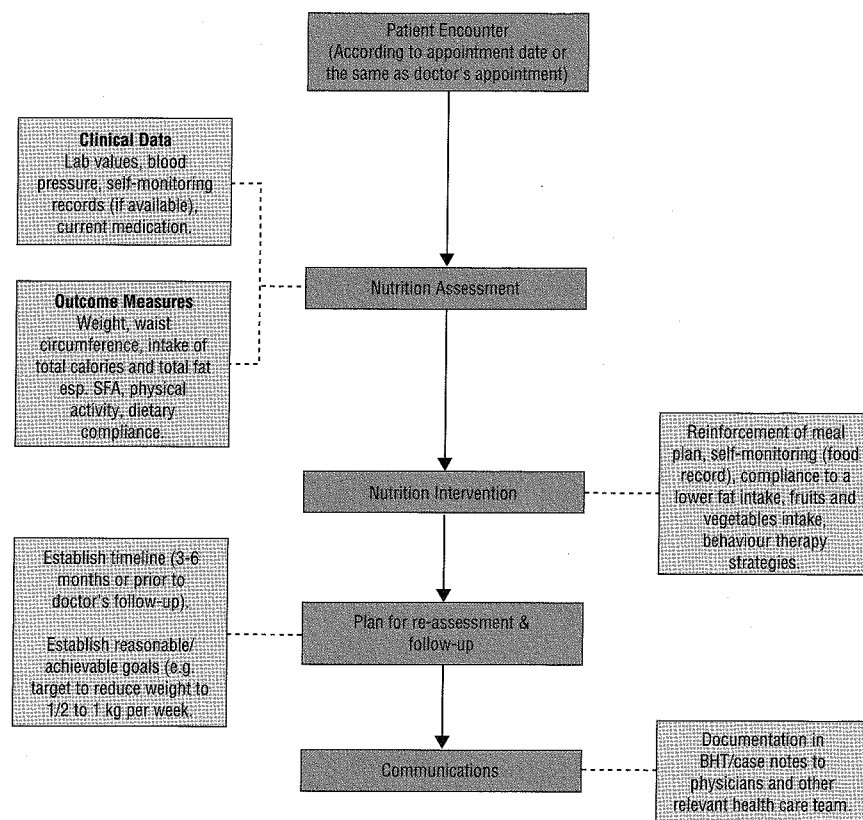
Subjective Information (What the patient reports)	• Date/time and reason for referral. • Age, gender, occupation, social history, alcohol intake and smoking habits. • Diagnosis, number of years with current disease, family history of disease. • Historical data include medical, psychological, social-economic, environmental factors e.g. previous nutritional problem, patients' lifestyle, physical activity level. • Patient comments about prescribed diet if any - knowledge on dietary management for hyperlipidemia, compliance to diet prescription. • Contraindicated foods - food allergies or intolerance
Objective Information (What is observed through the counseling and diagnostic tests)	• Anthropometric data - measurement of weight, height, BMI, waist circumference, calculation of ideal body weight and recommended weight. • Biochemistry results - Fasting Serum Lipid, Liver Function Test. Fasting/random blood glucose, HbA_{1c}, renal profile if applicable. • Clinical findings - blood pressure, medications taken and duration. • Diet history, eating out frequency, fat and cholesterol intake, alcohol intake. • Impaired ability to feed self /eat - any physical disability that limit patient's ability to feed self.
Assessment (The diagnosis based on interpretation and categorization of data; patient responses to health problems)	• Assessment of patient medical record, clinical data, anthropometric measurements, diet history and physical activity. • Patient's response, knowledge and attitude/behavior changes towards the disease and diet. • Assessment of carers' attitude & understanding of the prescribed diet. • Determination of nutritional requirement and recommendation.
Plan (Includes diagnostic, therapeutic, patient education, future actions and the evaluation of these actions)	• Establishment of nutrition care plan/recommendation including nutritional goals/ objectives, and weight reduction essential to the treatment of the nutritional problem. • Implementation, monitoring and revision of nutrition care plan. • Individualized treatment goals • Exercise/ physical activities frequency. • Follow-up plans and reassessments (future visit).

NUTRITION CARE PROCESS: 1ST VISIT



Patient 1st visit / encounter.
Length of contact: At least 45 minutes

NUTRITION CARE PROCESS: FOLLOW-UP VISIT



Patient Follow-up visit / encounter.
Length of contact: Within 30 minutes

SPECIAL TOPICS

DIFFERENT TYPES OF DYSLIPIDEMIAS

Randomized clinical trials generally have not focused on specific dyslipidemias. Yet these disorders are common enough to deserve specific attention in ATP III. Recommendations are based on ATP III.

Clinical condition	Clinical Features	Therapeutic Recommendations
Familial hypercholesterolemia (heterozygous)	- Prevalence 1 in 500 persons - Detectable at birth or shortly after - TC eventually rise to 9.0 - 12.0 mmol/L - LDL-C is twice of normal readings.	- Therapeutic lifestyle changes (TLC) - Generally drug therapy required as TLC is not effective alone.
Familial hypercholesterolemia (homozygous)	- Prevalence 1 in 1,000,000 persons - TC commonly at 18.0 - 31.0 mmol/L - LDL-C is 4 fold increase of the normal readings.	- Resistant to TLC and cholesterol lowering drugs. - Commercial techniques for removal of VLDL and LDL from the plasma (plasmapheresis) are Heparin-Induced Extracorporeal Lipoprotein Precipitation (HELP) and Dextran Sulphate Cellulose Absorbent. Such treatments must be performed every 1 - 3 weeks depending on the clinical state of the patient, in order to promote xanthoma regression and retard atheroma formation.
Familial defective apolipoprotein B-100 (FDB)	- Autosomal dominant causes elevated LDL - Prevalence 1 in 700 to 1000 persons. - LDL-C = 1.5 - 2.0 times more of normal readings.	- TLC initiation - All LDL-lowering drug therapy are effective
Polygenic hypercholesterolemia	- Prevalence 1 in 10 to 20 persons pending on age - LDL-C > 4.9 mmol/L	- TLC initiation - Drug therapy after dietary therapy

Elevated Triglycerides

Clinical condition	Clinical Features	Therapeutic Recommendations
Borderline High Triglycerides (1.7 - 2.2 mmol/L)	- Elevated small LDL particles. - Low HDL cholesterol - Elevated BP - Insulin resistance and glucose intolerance	TLC to achieve LDL-C goal. Followed-by: - Body weight control - Regular physical activity - Smoking cessation - Restriction of alcohol - Avoid high carbohydrate intake (> 60 % of total energy intake)
High Triglycerides (2.3 - 5.6 mmol/L)	Similar as borderline high triglycerides	- TLC to achieve LDL-C goal - Non HDL-cholesterol goal to be 0.8 mmol/L higher than LDL-C goal <i>To calculate non HDL-C</i> $= TC - HDL-C$, or $= LDL-C + VLDL$
Familial combined hyperlipidaemia	- TC = 6.5 - 9 mmol/L - TG = 2.3 - 5.6 mmol/L - May exhibit high apo B-100	
Familial hypertriglyceridemia	- No increased premature risk of CHD - Lower LDL-C	
Familial dysbetalipoproteinemia	Isoform variation of apolipoprotein E	
Very high triglycerides (> 5.6 mmol/L)	Metabolic syndrome, DM	- TG lowering to prevent pancreatitis - Very low fat diet if TG > 11.3 mmol/L (< 15 % of total energy intake as fat) - MCT if TG > 11.3 mmol/L to replace long chain TG - Weight reduction/ physical activity - Fish oils

Low HDL cholesterol (without hypertriglyceridemia)

Clinical condition	Clinical Features	Therapeutic Recommendations
Low HDL cholesterol (< 1.03 mmol/L)	- TG < 1.7 mmol/L - If TG is borderline high, HDL-C will fall	- Weight reduction: 5 - 20 % increase in HDL-C - Physical activity: 5 - 30 % increase in HDL-C - Smoking cessation: 5 % increase in HDL-C

OMEGA-3 FATTY ACIDS

Omega-3 fatty acids are a special class of polyunsaturated fatty acids that are essential in the diet, as they cannot be made from other forms of fats in the body. The principle forms of Omega 3s that the body needs are alpha-linolenic acid (ALA), docosahexanoic acid (DHA) and eicosapentanoic acid (EPA). Although the body cannot make Omega-3 fatty acids, it is able to convert them from one form of Omega-3 fatty acids to another (Kris-Etherton et al, 2002).

While the precise mechanisms of Omega-3 fatty acids' involvement in maintaining cardiovascular health are not precisely understood (ATP III), there is much evidence to show that these fatty acids are essential in maintaining normal cardiovascular function.

Evidence for the beneficial effects of Omega-3 fatty acids in cardiovascular health is provided by several sets of research. Omega 3 fatty acids supplementation in the diet of people with high triglyceride levels results in significant lowering of triglycerides (Albert et al, 2002). Triglycerides are a fatty substance in the blood that could lead to heart disease if present in excessive amounts. In particular high triglyceride levels are known to thicken the blood, slowing flow. When levels are elevated, the level of HDL or 'good' cholesterol is reduced.

American Heart Association in year 2002 (Kris-Etherton et al, 2002) has made the following recommendation:

Population	Recommendation
Patients without documented CHD	Eat a variety of (preferably oily) fish at least twice a week. Include oils & foods rich in alpha-linolenic acid (flaxseed, canola and soyabean oils; flaxseed and walnuts).
Patients with documented CHD	Consume ~1g of EPA+DHA per day, preferably from oily fish. EPA+DHA supplements could be considered in consultation with the physician.
Patients needing triglycerides lowering	Two to four grams of EPA+DHA per day provided as capsules under a physician's care.

Evidence Statement:

The mechanism whereby omega 3 fatty acids might reduce coronary events is unknown and maybe multiple. Prospective data and clinical trial evidence in secondary CHD prevention suggest that higher intakes of omega 3 fatty acids reduce risk for coronary events or coronary mortality.

Recommendation:

Higher dietary intakes of omega 3 fatty acids in the form of fatty fish or vegetable oils are an option for reducing risk for CHD. This recommendation is optimal because the strength of the evidence is only moderate at present. ATP III supports the American Heart Association recommendation that fish be included as part of a CHD risk reduction diet. Fish in general is low in saturated fat and may contain some cardio protective omega 3 fatty acids. However, a dietary recommendation for a specific amount of omega 3 fatty acid is not being made.

SOY PRODUCTS

The Food and Drug Administration (FDA) approved a health claim statement that 25 g of soya protein per day as part of a diet low in saturated fat and cholesterol may reduce the risk of heart disease (American Dietetic Assc, 2001).

The specific processing of the soya bean determines the characteristic of soya protein such as the content of isoflavones, fibre and saponins (ATP III).

Sources of soya protein include soya milk, soyabeans, tofu, soya-based meat and cheese substitutes, and alfalfa sprouts (American Dietetic Assc, 2001).

Evidence Statement:

High intake of soya protein can cause small reduction in LDL cholesterol levels especially when it replaces animal food products (ATP III).

Recommendation:

Food sources containing soya protein are acceptable as replacement for animal food product containing animal fats (ATP III).

GARLIC

Garlic belongs to the Allium family. It contains upward of 200 phytochemical compounds, including diallyl disulfide, thiols, and allylic sulfides. Garlic also contains nutrients such as vitamins A and C, and selenium.

Phytochemical compounds are biologically active plant molecules, which are produced by plants to protect themselves from ultraviolet radiation, soil and air pollutants, oxidation, viruses, bacteria, fungi, insects, and animals.

Most of the phytochemical compounds happen to be antioxidants (Tribble, 1999), which protect healthy molecules from oxidation by free radicals. In other words, these compounds also prevent lipid oxidation in human bodies and therefore prevent or slow down the progress of atherosclerosis (ATP III; Tribble, 1999; British Nutrition Foundation, 1997).

Evidence Statement:

Oxidative stress and LDL oxidation appear to be involved in atherogenesis. However, clinical trials to date have failed to demonstrate that supplementation of the diet with antioxidants will reduce risk for cardiovascular disease (ATP III).

Recommendation:

It is not recommended that supplementation with concentrated or purified preparation as a widespread policy for cardiovascular disease prevention, as long-term safety and efficacy in a variety of populations had not been demonstrated. In view of these findings, the most prudent and scientifically supportable recommendation for the general population is to consume a balanced diet with emphasis on antioxidant-rich fruits and vegetables and whole grains (Tribble, 1999; British Nutrition Foundation, 1997).

PALM OIL

Palm oil contains an equal proportion of saturated and unsaturated fatty acids, with about 44% palmitic acid, 5 % stearic acid, 40 % oleic acid, 10 % linolenic acid (both polyunsaturated).

Palm oil and palm olein must be distinguished clearly from palm kernel oil and coconut oil, because they have a very different fatty acid composition, with an appreciably lower level of saturated components, and no significant content of capric acids (C 10:0), lauric acids (C 12:0) or myristic acids (C 14:0). In general SFA tends to elevate blood cholesterol in all lipoprotein fractions (i.e., LDL and HDL cholesterol) when substituted for carbohydrate or other fatty acids. There is a dose response relationship between SFA and LDL-C (Van Horn and Ernt, 2001). The most hypercholesterolemic or atherogenic SFAs are lauric (C12:0), myristic (C14:0) and palmitic (C16:0) acids. Myristic is the most potent, followed by palmitic and then lauric acid. An SFA stearic acid (C18:0) has no effect on blood lipoproteins and is considered neutral, like carbohydrates (Kris - Etherton and Yu 1997) Palmitic acid is the most prevalent hypercholesterolemic SFA in the American diet. Though present in plant sources, most dietary palmitate comes from animal foods. Myristic acid is found mostly in butterfat, coconut and palm kernel oils which is more prevalent in the Malaysian diet. Lauric acids, the only medium chain in SFA, is found in palm kernel and coconut oils. Of all the added fats, the most hypercholesterolemic are palm kernel, coconut oils, lard and butter

Recent controlled human studies in Europe, the USA and Asia have confirmed that serum cholesterol does not increase when palm oil is used to replace major parts of fats in a traditional diet. Studies have shown that palm oil can reduce the total serum cholesterol level in individuals with hypercholesterolemia but it tends to have minimal effects in normocholesterolemic individuals (Augustine, 1994).

Recommendation:

It is recommended to maximize the potential of palm oil by having a mixture of polyunsaturated fatty acid oils in the ratio of 1: 2

HOMOCYSTEINE

The homocysteine theory was first presented by Dr Kilmer from Harvard University in 1969 but it was not well accepted by his peers. Much research has been done on homocysteine which is now proven as an independent risk factor for heart disease and it is even known as the 'new cholesterol'.

Homocysteine is a metabolite of the amino acid call methionine. Methionine is an amino acid derived from dietary protein such as meat, eggs or milk. The body has a mechanism of maintaining safe level of homocysteine. This requires folic acid, vitamin B6 and vitamin B12 (Kang et al, 1987; Robinson et al, 1998; Selhub et al, 1993). With the help of folic acid and vitamin B12, homocysteine is converted back into methionine. Vitamin B6 on the other hand helps to convert homocysteine into harmless amino acid cysteine that is excreted in the urine. Excess methionine in the body is converted to homocysteine. When homocysteine level rises, it begins to damage cells and tissues of arteries, and stimulate the build-up of plaque, which leads to heart disease.

There are 3 proposed mechanism involved:

1. Homocysteine is a growth promoter of smooth muscle cells.
2. High levels of blood homocysteine triggers blood clotting.
3. Homocysteine damages cell lining of the arteries (endothelium).

Observations in 80 clinical and epidemiological studies have suggested that elevated homocysteine is a risk factor for atherosclerotic vascular disease and for arterial and venous thromboembolism (Refsum et al, 1988). The strongest evidence for a relationship between homocysteine and cardiovascular disease was provided by 6 prospective studies (Malinov et al, 1997; Goyette, 1994; Frosst et al, 1995; Kang et al, 1991; Ou et al, 1998; Food and Nutrition Board) with follow ups from 1.4 to 12.8 years on 830 cases and 1872 controls. It is premature to conclude that homocysteine levels are predictive of the development of cardiovascular disease because of absence of controlled clinical intervention trials.

Evidence statement:

According to the Institute of Medicine, the RDA folate for adults is 400 microgram per day and the upper limit is 100 microgram per day. There are no published randomized clinical trials to show whether lowering homocysteine levels through dietary intake or supplements of folate and other B vitamins will reduce the risk for CHD.

Recommendation:

ATP endorsed the Institute of Medicine RDA for dietary folate namely 400 microgram per day. Folate should be consumed largely from dietary sources such as green leafy vegetables, orange, beans, lentils (ATP III).

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APPENDIX

APPENDIX 1: ESSENTIAL PATIENT EDUCATION TOPICS FOR PATIENTS WITH CVD.

Principles of MNT that need to be taught to and understood by the patients.	Education Topics
Food preparation	Uses of healthier cooking techniques to modify fat intake. Healthier cooking techniques using minimum oil include steam, boil, grill, baked, roast, stew and microwave.
Recipe modification	Modifies recipes to lower saturated fat, trans fats. Substitute: • Santan with low fat, skimmed milk, yoghurt or unsweetened soy milk • Butter with soft margarine • For baking, 1 egg yolk can be substituted with 2 egg whites
Eating out	Select food choices appropriately Add more vegetables or beans to main entrees to increase fiber content For example, choose: • Clear vegetable soups instead of cream soups • Laksa assam instead of laksa lemak • Soup-based dishes instead of fried dishes • Chapatti, thosai, naan or idli instead of roti canai, roti telur, roti bom, roti tisu, murtabak • White rice instead of oily flavoured rice Traveling schedules such as attending conferences and business meetings should be addressed.
Shopping	• Make a list of groceries. • Read food labels and nutrition claims. • Information on low fat, low-cholesterol and low-salt foods.
Snacking habits	Choose low fat snacks and reduce the habits to the minimal intake. For example, choose: • Fresh fruits instead of sweet dessert • Popiah basah instead of fried popiah • Wholemeal sandwich or plain buns instead of fried kuih • Wholemeal biscuits instead of assorted cookies or kuih • Low fat yoghurt instead of ice cream

APPENDIX 1: ESSENTIAL PATIENT EDUCATION TOPICS FOR PATIENTS WITH CVD.

Principles of MNT that need to be taught to and understood by the patients.	Education Topics
Exercise	- Patients should be encouraged to engage in a variety of physical activities and to progressively increase their activity as tolerated. - Recommendation of 30 minutes or more of moderate-intensity physical activity such as brisk walking on most, and preferably all, days of the week. - Recommendation to lengthen duration of activity as appropriate beyond 15 minutes and to as much as 45 to 60 minutes per session is valuable for increasing caloric expenditure, but lesser intensity should be used. This is associated with improvement in a number of risk factors including obesity, lipid abnormalities, hypertension and elevated blood glucose. Post Coronary Artery Bypass Graft (CABG): - Exercise plays an important part to aid recovery. Low level activities such as slow walking usually acceptable 48 hours after surgery. - Chest and leg wound usually require 4 - 6 weeks for healing. Upper body exercises that cause sternal tension should be avoided for up to 3 months after surgery.
Alcohol intake	Limit to 1 - 2 drinks per day.
Smoking cessation	- Verbalize importance of smoking cessation. - To quit smoking. - Passive smoking also increases the risk of heart disease.
Behavioral modification techniques	Patients' readiness to change.

APPENDIX 2: RISK FACTORS FOR CORONARY HEART DISEASE

Table A: ATP III Classification of Total Cholesterol and LDL-C

Total Cholesterol (mmol/L & mg/dL)		LDL-cholesterol (mmol/L & mg/dL)	
		< 2.6 mmol/L < 100 mg/dL	Optimal
< 5.2 mmol/L < 200 mg/dL	Desirable	2.6 - 3.3 mmol/L 100 - 129 mg/dL	Near Optimal / above optimal
5.2 - 6.2 mmol/L 200 - 239 mg/dL	Borderline High	3.4 - 4.1 mmol/L 130 - 159 mg/dL	Borderline High
≥ 6.2 mmol/L ≥ 240 mg/dL	High	4.1 - 4.9 mmol/L 160 - 189 mg/dL	High
		≥ 4.9 mmol/L ≥ 190 mg/dL	Very High

Table B: Clinical Identification of the Metabolic Syndrome*

Risk Factors	Defining Level
Abdominal Obesity	Waist Circumference†
Men Women	> 40 in (> 102 cm) > 35 in (> 88 cm)
Triglycerides	> 1.69 mmol/L (150 mg/dL)
HDL cholesterol	
Men Women	< 1.04 mmol/L (40 mg/dL) < 1.29 mmol/L (50 mg/dL)
Blood pressure	> 130/ 85 mmHg
Fasting glucose	> 6.1 mmol/L (110 mg/dL)

* The ATP III panel did not find adequate evidence to recommend routine measurement of insulin resistance (e.g. plasma insulin), proinflammatory state (e.g. high-sensitivity C-reactive protein), or prothrombotic state (e.g. fibrinogen or PAI-1) in the diagnosis of the metabolic syndrome.

† Some male persons can develop multiple metabolic risk factors when the waist circumference is only marginally increased e.g. 94 - 102 cm (37 - 39 in). Such person may only have a strong genetic contribution to insulin resistance. They should benefit from changes in life habits, similarly to men with categorical increase in waist circumference.

APPENDIX 3: SATURATED FAT AND CHOLESTEROL FOOD LIST

Portion	Food	Saturated Fat (g)	% Daily total*	Dietary Cholesterol (mg)	% Daily total**
Milk Products					
1 Tbsp.	Butter	7.2	33	31	10
1 oz. ***	Cheese, American	4.4	20	18	6
1 oz.	Cheese, cheddar	6.0	5	30	10
1 oz.	Cheese, mozzarella, part-skim	3.1	14	15	5
1 Tbsp.	Margarine, hard stick	2.1	10	0	0
1 Tbsp.	Margarine, soft tub	1.5	7	0	0
8 oz.	Milk, nonfat / skimmed	0.0	0	5	2
8 oz.	Milk, reduced fat, 2 % milk fat	2.9	13	20	7
8 oz.	Milk, whole, 3.7 % milk fat	5.6	26	34	11
Eggs					
Small	Egg, fresh	1.1	5	157	52
Medium	Egg, fresh	1.4	6	187	62
Large	Egg, fresh	1.6	7	213	71
5 whole	Quail's egg	n/a	n/a	188	63
Meats					
3 oz.	Beef, ground, lean, 21 % fat	5.9	27	86	29
3 oz.	Beef, ground, regular, 27 % fat	6.5	30	86	29
3 oz.	Beef, t-bone, meat only	3.1	14	50	17
3 oz.	Beef, tenderloin, meat only	3.1	14	73	24
3 oz.	Ham, regular, 11 % fat	2.9	13	49	16
3 oz.	Ham, extra lean, 5 % fat	1.4	6	40	13
1 item	Hot dog, beef, 1 = 45 g	5.4	25	27	9
3 oz.	Lamb, loin chop, lean only	3.2	15	74	25
3 oz.	Pork, loin chop, meat only	3.0	14	69	23
Poultry					
3 oz.	Chicken, breast, meat only	0.9	4	73	24
3 oz.	Chicken, breast, meat and skin	1.8	8	71	24
3 oz.	Chicken, thigh, meat only	2.5	11	80	27
Fish and Seafood					
3 oz.	Clams	0.0	0	57	19
3 oz.	Cod	0.0	0	45	15
3 oz.	Oysters	0.9	4	85	28
3 oz.	Lobster	0.0	0	69	23
3 oz.	Salmon (varies by species and whether farm-raised or wild)	0.0-2.7	0-12	47-81	16-27
3 oz.	Scallops	0.0	0	28	9
3 oz.	Shrimp	0.0	0	166	55

APPENDIX 3: SATURATED FAT AND CHOLESTEROL FOOD LIST

Portion	Food	Saturated Fat (g)	% Daily total*	Dietary Cholesterol (mg)	% Daily total**
Fish and Seafood					
3 oz.	Tuna, canned in water, light	0.0	0	26	9
3 oz.	Tuna, fresh	0.0	0	50	17
3 oz.	Tenggiri	n/a	n/a	3	1
3 oz.	Ikan kembung	n/a	n/a	55	18
3 oz.	Ikan merah	n/a	n/a	14	5
3 oz.	Bawal hitam	n/a	n/a	51	17
3 oz.	Bawal putih	n/a	n/a	27	9
3 oz.	Ikan selar	n/a	n/a	47	16
3 oz.	Ikan cencaru	n/a	n/a	46	19
1/2 cup	Ikan bilis, without head	n/a	n/a	37	12
	Ikan terubuk	n/a	n/a	74	35
3 oz.	Sardine, canned in tomato sauce	n/a	n/a	107	36
1 oz.	Fish roe (telur ikan)	n/a	n/a	98	33
3 oz.	Squid	n/a	n/a	90	30
Snack Foods					
1 oz.	Crackers, snack	1.0	5	0	0
Small	French fries, fast-food	3.3	15	0	0
Med	French fries, fast-food	5.1	23	0	0
Large	French fries, fast-food	6.5	30	0	0
1 oz.	Potato chips	2.3	11	0	0
1 pc, 115 g	Kuih apam balik	5	2	0	0
1 pc, 30 g	Kuih koci	8	4	0	0
1 bowl, 240 g	Bubur cha cha	4	1	0	0
1 bowl, 240 g	Cendol	8	3	0	0
1 tbsp	Kaya	0.3	0	6	2

* Based on a 2,000 calorie per day diet.

** Based on a recommended cholesterol limit of less than 300 mg per day

*** 1 oz is equivalent to 30 g

Reference:

1. AHA Dietary Guidelines. Circulation 2000; 102:2284-2299.

2. Nutrient Composition of Malaysian Food 4th edition. Institute of Medical Research, 1998.

3. Keeping Fat in Check, Food & Nutrition Department, Ministry of Health Singapore, 1996.

APPENDIX 4: SAMPLE MENU - LOW CHOLESTEROL & LOW FAT FOR 1800 CALORIES

Breakfast

- High fiber bread (2 slices, 50 g) with soft monounsaturated oil margarine (1 dsp, 10 g)
- Unsweetened soybean milk (250 ml)

Lunch

- Rice (1 bowl, 160 g)
- Singgang ikan merah (1 piece, 120 g)
- Stir fried cabbage (1/2 cup, 70 g)
- Stir fried kangkung (1/2 cup, 70 g)
- Polyunsaturated oil (1 dsp, 10 g)
- Guava (1 slice, 130 g)

Afternoon snack

- Tea with low fat milk (1/2 cup, 125 ml) and sugar (1 tsp, 5 g)
- Sardine sandwich (2 piece, 56 g)

Dinner

- Capati (1 piece, 100 g)
- Chicken tikka
 - Chicken (1 piece, 120 g)
- Dhal (1/2 cup, 80 g)
- Saag
 - Spinach (1/2 cup, 60 g)
- Cucumber and carrot raita (1 cup, 140 g)
- Polyunsaturated oil (1 dsp, 10 g)
- Papaya (1 slice, 160 g)

Nutritional Analysis:

Energy	1747 kcal	Fat	57 g (29%)
Protein	80 g (18%)	• SFA	8 g (4 %)
Carbohydrate	238 g (53%)	• PUFA	14 g (7 %)
Dietary Fiber	24.5 g	• MUFA	18 g (9 %)
Cholesterol	138 mg		

APPENDIX 5: MEDICATION

Anti Lipid Drug Classes	Description	Side Effects
HMG-CoA reductase inhibitors (statins) • Lovastatin • Pravastatin • Simvastatin • Fluvastatin • Atorvastatin	To lower LDL-C. Reduces risk for CHD and stroke	Myopathy, increased liver enzymes Contraindication: Absolute • Active / chronic liver disease Relative • Concomitant use of certain drugs*
Fibric-Acid Derivatives (fibrates) • Gemfibrozil • Fenofibrate • Clofibrate	Hypertriglyceridemia, atherogenic dyslipidemia. Clinical trial evidence of CHD risk reduction.	Dyspepsia, gallstones, myopathy Contraindication: Absolute • Severe hepatic disease • Severe renal disease
Bile Acid Sequestrants (resins) • Cholestyramine • Colestipol • Colesevelam	To lower LDL-C. Clinical trial evidence of CHD risk reduction.	GI distress, constipation, decreased absorption of certain drugs** Contraindication: Absolute • Dysbetalipoproteinemia • TG > 4.5 mmol/l Relative • TG > 2.3 mmol/l
Nicotinic Acid (Niacin) • Crystalline nicotinic acid • Sustained release nicotinic acid • Extended release nicotinic acid	Useful in most lipid and lipoprotein abnormalities. Clinical trial evidence of CHD risk reduction.	Flushing, hyperglycemia, hyperuricemia (or gout), upper GIT distress, hepatotoxicity Contraindication: Absolute • Chronic liver disease • Severe gout Relative • Diabetes (high doses only) • Hyperuricemia • Peptic ulcer disease
Cholesterol Absorption Inhibitors***		Headache, abdominal pain, diarrhea

* cyclosporin, macrolide antibiotics, various anti fungal agents and cytochrome P-450 inhibitors (fibrates and nicotinic acid should be used with appropriate caution)

** Paracetamol, NSAIDs, anticoagulant, valproate, digitalis, thiazides, thyroxine, raloxifene, propranolol, tricyclic antidepressants

*** usually used in combination with statins

Reference: Clinical Practice Guideline on Dyslipidemia, 2004.